

Head Impact Detection Using Kinematic Data

1. Problem Statement

Head impacts are a major concern in contact sports such as American football. Detecting impacts accurately can help reduce false alarms and identify potential risks. This project compares four machine learning models—HIKNet, RecursiveNet, Support Vector Machine (SVM), and XGBoost—to classify kinematic signals as impact or non-impact (false positive).

2. Dataset Details

The original research paper used 527 real head-impact recordings collected using instrumented mouthguards from Stanford football players. Each recording contains six channels: linear acceleration (X, Y, Z) and angular velocity (X, Y, Z).

Since the original dataset was unavailable, a synthetic dataset was generated containing 527 samples, 199 time points per sample, and six channels. It includes 264 impact-labelled samples and 263 non-impact-labelled samples. The data was divided into 369 training samples and 158 evaluation samples, with 79 samples per class in the evaluation set.

Limitation: The synthetic signals and labels are not real-world measurements or verified impact annotations. Therefore, the results demonstrate model performance on synthetic data only.

3. Approach and Methodology

The project compares deep learning models using raw signals with traditional machine learning models using extracted features.

- **HIKNet:** Uses one-dimensional and two-dimensional convolutional layers to learn patterns directly from raw signals.
- **RecursiveNet:** Uses a deeper convolutional architecture with recursive connections to learn signal patterns.
- **SVM:** Uses 176 engineered features describing peak values, signal duration, and frequency information. The features are standardized before classification.
- **XGBoost:** Uses the same 176 features as SVM and combines decision trees for classification.

The models were evaluated on the same held-out test set using accuracy, precision, specificity, sensitivity, ROC-AUC, and PR-AUC.

4. Implementation and Results

The project was implemented in Python, covering synthetic data generation, feature extraction, model training, and evaluation. A fixed random seed was used for reproducibility.

Model	Accuracy	Sensitivity	ROC-AUC
HIKNet	91.14%	92.41%	0.946
RecursiveNet	84.81%	83.54%	0.917
SVM	79.11%	68.35%	0.905
XGBoost	89.24%	88.61%	0.933

HIKNet achieved the highest accuracy, sensitivity, ROC-AUC, and PR-AUC. XGBoost achieved the highest precision at 89.74% and performed better than SVM in accuracy and sensitivity. These results indicate that HIKNet performed best overall on the synthetic dataset.

5. Conclusion

This project compared four machine learning models for classifying head-impact-like kinematic signals. HIKNet achieved the best overall performance, while XGBoost was a competitive feature-based approach. However, the results are limited by the use of synthetic data and a single evaluation split. Future work should use real, labelled recordings and evaluate models across multiple data splits or through cross-validation to assess their reliability.